

Question ID cb8f449f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: cb8f449f

$$\frac{1}{2}y = 4$$

$$x - \frac{1}{2}y = 2$$

The system of equations above has solution (x, y). What is the value of x ?

- A. 3
- B. $\frac{7}{2}$
- C. 4
- D. 6

ID: cb8f449f Answer

Correct Answer:
D

Rationale

Choice D is correct. Adding the corresponding sides of the two equations eliminates y and yields $x = 6$, as shown.

$$\frac{1}{2}y = 4$$

$$x - \frac{1}{2}y = 2$$

$$\hline x + 0 = 6$$

If (x, y) is a solution to the system, then (x, y) satisfies both equations in the system and any equation derived from them. Therefore, $x = 6$.

Choices A, B, and C are incorrect and may be the result of errors when solving the system.

Question Difficulty:
Medium

Question ID bf4a8b6a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: bf4a8b6a

A company that provides whale-watching tours takes groups of **21** people at a time. The company’s revenue is **80** dollars per adult and **60** dollars per child. If the company’s revenue for one group consisting of adults and children was **1,440** dollars, how many people in the group were children?

- A. 3
- B. 9
- C. 12
- D. 18

ID: bf4a8b6a Answer

Correct Answer:
C

Rationale

Choice C is correct. Let x represent the number of children in a whale-watching tour group. Let y represent the number of adults in this group. Because it's given that **21** people are in a group and the group consists of adults and children, it must be true that $x + y = 21$. Since the company's revenue is **60** dollars per child, the total revenue from x children in this group was $60x$ dollars. Since the company's revenue is **80** dollars per adult, the total revenue from y adults in this group was $80y$ dollars. Because it's given that the total revenue for this group was **1,440** dollars, it must be true that $60x + 80y = 1,440$. The equations $x + y = 21$ and $60x + 80y = 1,440$ form a linear system of equations that can be solved to find the value of x , which represents the number of children in the group, using the elimination method. Multiplying both sides of the equation $x + y = 21$ by **80** yields $80x + 80y = 1,680$. Subtracting $60x + 80y = 1,440$ from $80x + 80y = 1,680$ yields $(80x + 80y) - (60x + 80y) = 1,680 - 1,440$, which is equivalent to $80x - 60x + 80y - 80y = 240$, or $20x = 240$. Dividing both sides of this equation by **20** yields $x = 12$. Therefore, **12** people in the group were children.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of adults in the group, not the number of children in the group.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID e53688cb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: e53688cb

$$\begin{aligned}x + 3y &= 29 \\ 3y &= 11\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

ID: e53688cb Answer

Correct Answer:
18

Rationale

The correct answer is 18. It's given by the second equation in the system that $3y = 11$. Substituting 11 for $3y$ in the first equation in the system, $x + 3y = 29$, yields $x + 11 = 29$. Subtracting 11 from both sides of this equation yields $x = 18$.

Question Difficulty:
Medium

Question ID 65833256

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 65833256

$$\begin{aligned}y &= 6x + 16 \\ -7x - y &= 36\end{aligned}$$

What is the solution (x, y) to the given system of equations?

- A. $(-4, -8)$
- B. $(-\frac{20}{13}, -\frac{80}{13})$
- C. $(4, 40)$
- D. $(20, 136)$

ID: 65833256 Answer

Correct Answer:

A

Rationale

Choice A is correct. The given system of linear equations can be solved by the substitution method. The first equation in the given system of equations defines y as $6x + 16$. Substituting $6x + 16$ for y in the second equation of the given system of equations yields $-7x - (6x + 16) = 36$. Applying the distributive property on the left-hand side of this equation yields $-7x - 6x - 16 = 36$, or $-13x - 16 = 36$. Adding 16 to both sides of this equation yields $-13x = 52$. Dividing both sides of this equation by -13 yields $x = -4$. Substituting -4 for x in the first equation of the given system of equations, $y = 6x + 16$, yields $y = 6(-4) + 16$, or $y = -8$. Therefore, the solution (x, y) to the given system of equations is $(-4, -8)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Medium

Question ID b544a348

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: b544a348

$$\begin{aligned}5x + 3y &= 38 \\ x + 3y &= 10\end{aligned}$$

In the solution (x, y) to the system of equations above, what is the value of x ?

ID: b544a348 Answer

Rationale

The correct answer is 7. Subtracting the second equation from the first equation eliminates the variable y .

$$\begin{array}{rcl}5x + 3y & = & 38 \\ -(x + 3y & = & 10) \\ \hline 4x & = & 28\end{array}$$

Dividing both sides of the resulting equation by 4 yields $x = 7$.

Question Difficulty:
Medium

Question ID e3bbde69

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: e3bbde69

$$\begin{aligned}8x + y &= 5 \\ y &= 9x + 1\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. -6
- B. $\frac{4}{17}$
- C. $\frac{6}{17}$
- D. 4

ID: e3bbde69 Answer

Correct Answer:
B

Rationale

Choice B is correct. The second equation in the given system is $y = 9x + 1$. Substituting $9x + 1$ for y in the first equation in the given system yields $8x + 9x + 1 = 5$, which is equivalent to $17x + 1 = 5$. Subtracting 1 from both sides of this equation yields $17x = 4$. Dividing both sides of this equation by 17 yields $x = \frac{4}{17}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 686b7cad

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 686b7cad

A proposal for a new library was included on an election ballot. A radio show stated that **3** times as many people voted in favor of the proposal as people who voted against it. A social media post reported that **15,000** more people voted in favor of the proposal than voted against it. Based on these data, how many people voted against the proposal?

- A. 7,500
- B. 15,000
- C. 22,500
- D. 45,000

ID: 686b7cad Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that a radio show stated that **3** times as many people voted in favor of the proposal as people who voted against it. Let x represent the number of people who voted against the proposal. It follows that $3x$ is the number of people who voted in favor of the proposal and $3x - x$, or $2x$, is how many more people voted in favor of the proposal than voted against it. It's also given that a social media post reported that **15,000** more people voted in favor of the proposal than voted against it. Thus, $2x = 15,000$. Since $2x = 15,000$, the value of x must be half of **15,000**, or **7,500**. Therefore, **7,500** people voted against the proposal.

Choice B is incorrect. This is how many more people voted in favor of the proposal than voted against it, not the number of people who voted against the proposal.

Choice C is incorrect. This is the number of people who voted in favor of the proposal, not the number of people who voted against the proposal.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Medium

Question ID a94ed4e0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: a94ed4e0

$$\begin{aligned} 3y &= 4x + 17 \\ -3y &= 9x - 23 \end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $39x$?

- A. -18
- B. -6
- C. 6
- D. 18

ID: a94ed4e0 Answer

Correct Answer:
D

Rationale

Choice D is correct. Adding the second equation to the first equation in the given system of equations yields $3y - 3y = 4x + 9x + 17 - 23$, or $0 = 13x - 6$. Adding 6 to each side of this equation yields $6 = 13x$. Multiplying each side of this equation by 3 yields $18 = 39x$. Therefore, the value of $39x$ is 18 .

Choice A is incorrect. This is the value of $-39x$, not $39x$.

Choice B is incorrect. This is the value of $-13x$, not $39x$.

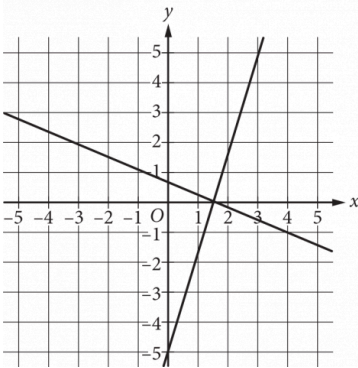
Choice C is incorrect. This is the value of $13x$, not $39x$.

Question Difficulty:
Medium

Question ID 2704399f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 2704399f



Which of the following systems of equations has the same solution as the system of equations graphed above?

- $y = 0$
- A. $x = \frac{3}{2}$
- $y = \frac{3}{2}$
- B. $x = 0$
- $y = 0$
- C. $x = 1$
- $y = 1$
- D. $x = 0$

ID: 2704399f Answer

Correct Answer:
A

Rationale

Choice A is correct. The solution to a system of equations is the coordinates of the intersection point of the graphs of the equations in the xy -plane. Based on the graph, the solution to the given system of equations is best approximated as $(\frac{3}{2}, 0)$. In the xy -plane, the graph of $y = 0$ is a horizontal line on which every y -coordinate is 0, and the graph of $x = \frac{3}{2}$ is a vertical line on which every x -coordinate is $\frac{3}{2}$. These graphs intersect at the point $(\frac{3}{2}, 0)$. Therefore, the system of equations in choice A has the same solution as the given system.

Choices B, C, and D are incorrect. If graphed in the xy-plane, these choices would intersect at the points $(0, \frac{3}{2})$, $(1, 0)$, and $(0, 1)$, respectively, not $(\frac{3}{2}, 0)$.

Question Difficulty:

Medium

Question ID 55447be2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 55447be2

$$\begin{aligned}y &= \frac{1}{3}x - 14 \\ y &= -x + 18\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

ID: 55447be2 Answer

Correct Answer:
24

Rationale

The correct answer is **24**. The given system of equations can be solved by the substitution method. The first equation in the given system of equations is $y = \frac{1}{3}x - 14$. Substituting $\frac{1}{3}x - 14$ for y in the second equation in the given system yields $\frac{1}{3}x - 14 = -x + 18$. Adding **14** to both sides of this equation yields $\frac{1}{3}x = -x + 32$. Adding x to both sides of this equation yields $\frac{4}{3}x = 32$. Multiplying both sides of this equation by $\frac{3}{4}$ yields $x = 24$.

Question Difficulty:
Medium

Question ID 5e422ff9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 5e422ff9

$$y = 2x - 3$$

$$3y = 5x$$

In the solution to the system of equations above, what is the value of y ?

- A. -15
- B. -9
- C. 9
- D. 15

ID: 5e422ff9 Answer

Correct Answer:
D

Rationale

Choice D is correct. Multiplying both sides of $y = 2x - 3$ by 5 results in $5y = 10x - 15$. Multiplying both sides of $3y = 5x$ by 2 results in $6y = 10x$. Subtracting the resulting equations yields $5y - 6y = (10x - 15) - (10x)$, which simplifies to $-y = -15$. Dividing both sides of $-y = -15$ by -1 results in $y = 15$.

Choices A and B are incorrect and may result from incorrectly subtracting the transformed equation. Choice C is incorrect and may result from finding the value of x instead of the value of y .

Question Difficulty:
Medium

Question ID b3c7ca1d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: b3c7ca1d

Which of the following systems of linear equations has no solution?

- A. $x = 3$
 $y = 5$
- B. $y = 6x + 6$
 $y = 5x + 6$
- C. $y = 16x + 3$
 $y = 16x + 19$
- D. $y = 5$
 $y = 5x + 5$

ID: b3c7ca1d Answer

Correct Answer:
C

Rationale

Choice C is correct. A system of two linear equations in two variables, x and y , has no solution if the graphs of the lines represented by the equations in the xy -plane are distinct and parallel. The graphs of two lines in the xy -plane represented by equations in slope-intercept form, $y = mx + b$, where m and b are constants, are parallel if their slopes, m , are the same and are distinct if their y -coordinates of the y -intercepts, b , are different. In the equations $y = 16x + 3$ and $y = 16x + 19$, the values of m are each **16**, and the values of b are **3** and **19**, respectively. Since the slopes of these lines are the same, and the y -coordinates of the y -intercepts are different, it follows that the system of linear equations in choice C has no solution.

Choice A is incorrect. The lines represented by the equations in this system are a vertical line and a horizontal line. Therefore, this system has a solution, **(3, 5)**, rather than no solution.

Choice B is incorrect. The two lines represented by these equations have different slopes and the same y -coordinate of the y -intercept. Therefore, this system has a solution, **(0, 6)**, rather than no solution.

Choice D is incorrect. The two lines represented by these equations are a horizontal line and a line with a slope of **5** that have the same y -coordinate of the y -intercept. Therefore, this system has a solution, **(0, 5)**, rather than no solution.

Question Difficulty:
Medium

Question ID e77a76ce

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: e77a76ce

Which of the following systems of linear equations has no solution?

- A. $y = 6x + 3$
 $y = 6x + 9$
- B. $y = 10$
 $y = 10x + 10$
- C. $y = 14x + 14$
 $y = 10x + 14$
- D. $x = 3$
 $y = 10$

ID: e77a76ce Answer

Correct Answer:
A

Rationale

Choice A is correct. A system of two linear equations in two variables, x and y , has no solution if the graphs of the lines represented by the equations in the xy -plane are distinct and parallel. The graphs of two lines in the xy -plane represented by equations in slope-intercept form, $y = mx + b$, where m and b are constants, are parallel if their slopes, m , are the same and are distinct if their y -coordinates of the y -intercepts, b , are different. In the equations $y = 6x + 3$ and $y = 6x + 9$, the values of m are each 6 , and the values of b are 3 and 9 , respectively. Since the slopes of these lines are the same and the y -coordinates of the y -intercepts are different, it follows that the system of linear equations in choice A has no solution.

Choice B is incorrect. The two lines represented by these equations are a horizontal line and a line with a slope of 10 that have the same y -coordinate of the y -intercept. Therefore, this system has a solution, $(0, 10)$, rather than no solution.

Choice C is incorrect. The two lines represented by these equations have different slopes and the same y -coordinate of the y -intercept. Therefore, this system has a solution, $(0, 14)$, rather than no solution.

Choice D is incorrect. The two lines represented by these equations are a vertical line and a horizontal line. Therefore, this system has a solution, $(3, 10)$, rather than no solution.

Question Difficulty:
Medium

Question ID dcc4886a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: dcc4886a

$y = \frac{2}{7}x + 3$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. If the second equation in the system is $y = mx + b$, where m and b are constants, what is the value of b ?

- A. -3
- B. $-\frac{1}{3}$
- C. $\frac{1}{3}$
- D. 3

ID: dcc4886a Answer

Correct Answer:
D

Rationale

Choice D is correct. It’s given that the system has infinitely many solutions. The graphs of two lines in the xy -plane represented by equations in slope-intercept form, $y = mx + b$, where m and b are constants, have infinitely many solutions if their slopes, m , are the same and if their y -coordinates of the y -intercepts, b , are also the same. The first equation in the given system is $y = \frac{2}{7}x + 3$. For this equation, the slope is $\frac{2}{7}$ and the y -coordinate of the y -intercept is 3 . If the second equation is in the form $y = mx + b$, then for the two equations to be equivalent, the values of m and b in the second equation must equal the corresponding values in the first equation. Therefore, the second equation must have a slope, m , of $\frac{2}{7}$, and a y -coordinate of the y -intercept, b , of 3 . Thus, the value of b is 3 .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty:
Medium

Question ID 4e400635

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 4e400635

$$\begin{aligned} 3x + 6 &= 4y \\ 3x + 4 &= 2y \end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of y ?

ID: 4e400635 Answer

Correct Answer:
1

Rationale

The correct answer is **1**. Subtracting the second equation from the first equation in the given system of equations yields $(3x - 3x) + (6 - 4) = 4y - 2y$, which is equivalent to $0 + 2 = 2y$, or $2 = 2y$. Dividing each side of this equation by **2** yields $1 = y$.

Question Difficulty:
Medium

Question ID 2875ba81

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 2875ba81

$$\begin{aligned}6x + 7y &= 28 \\ 2x + 2y &= 10\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of y ?

- A. -2
- B. 7
- C. 14
- D. 18

ID: 2875ba81 Answer

Correct Answer:
A

Rationale

Choice A is correct. The given system of linear equations can be solved by the elimination method. Multiplying each side of the second equation in the given system by 3 yields $(2x + 2y)(3) = (10)(3)$, or $6x + 6y = 30$. Subtracting this equation from the first equation in the given system yields $(6x + 7y) - (6x + 6y) = (28) - (30)$, which is equivalent to $(6x - 6x) + (7y - 6y) = 28 - 30$, or $y = -2$.

Choice B is incorrect. This is the value of x , not the value of y .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 6e50ce28

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 6e50ce28

The sum of a number x and 7 is twice as large as a number y . The number y is 3 less than the number x . Which system of equations describes this situation?

- A. $x + 7 = 2y$
 $y = x - 3$
- B. $x + 7 = 2y$
 $y = 3 - x$
- C. $2(x + 7) = y$
 $y = x - 3$
- D. $2(x + 7) = y$
 $y = 3 - x$

ID: 6e50ce28 Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that the sum of a number x and 7 is twice as large as a number y . This can be described by the equation $x + 7 = 2y$. It's also given that the number y is 3 less than the number x . This can be described by the equation $y = x - 3$. Therefore, the system consisting of the equations $x + 7 = 2y$ and $y = x - 3$ describes this situation.

Choice B is incorrect. The equation $y = 3 - x$ describes a situation where the number y is x less than 3 .

Choice C is incorrect. The equation $2(x + 7) = y$ describes a situation where the number y is twice the sum of a number x and 7 .

Choice D is incorrect. The equation $2(x + 7) = y$ describes a situation where the number y is twice the sum of a number x and 7 , and the equation $y = 3 - x$ describes a situation where a number y is x less than 3 .

Question Difficulty:
Medium

Question ID 9f6f96ff

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 9f6f96ff

A wire with a length of **106** inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is **6** more than **4** times the value of y . What is the value of x ?

- A. 25
- B. 28
- C. 56
- D. 86

ID: 9f6f96ff Answer

Correct Answer:

D

Rationale

Choice D is correct. It's given that a wire with a length of **106** inches is cut into two parts. It's also given that one part has a length of x inches and the other part has a length of y inches. This can be represented by the equation $x + y = 106$. It's also given that the value of x is **6** more than **4** times the value of y . This can be represented by the equation $x = 4y + 6$. Substituting $4y + 6$ for x in the equation $x + y = 106$ yields $4y + 6 + y = 106$, or $5y + 6 = 106$. Subtracting **6** from each side of this equation yields $5y = 100$. Dividing each side of this equation by **5** yields $y = 20$. Substituting **20** for y in the equation $x = 4y + 6$ yields $x = 4(20) + 6$, or $x = 86$.

Choice A is incorrect. This value represents less than half of the total length of **106** inches; however, x represents the length of the longer part of the wire, since it's given that the value of x is **6** more than **4** times the value of y .

Choice B is incorrect. This value represents less than half of the total length of **106** inches; however, x represents the length of the longer part of the wire, since it's given that the value of x is **6** more than **4** times the value of y .

Choice C is incorrect. This represents a part that is **6** more than the length of the other part, rather than **6** more than **4** times the length of the other part.

Question Difficulty:

Medium

Question ID 79784c23

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 79784c23

$$y = 6x + 3$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. Which equation could be the second equation in this system?

- A. $y = 2(6x) + 3$
- B. $y = 2(6x + 3)$
- C. $2(y) = 2(6x) + 3$
- D. $2(y) = 2(6x + 3)$

ID: 79784c23 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions when the two linear equations are equivalent. When one equation is a multiple of another equation, the two equations are equivalent. Multiplying each side of the given equation by 2 yields $2(y) = 2(6x + 3)$. Thus, $2(y) = 2(6x + 3)$ is equivalent to the given equation and could be the second equation in the system.

Choice A is incorrect. The system consisting of this equation and the given equation has one solution rather than infinitely many solutions.

Choice B is incorrect. The system consisting of this equation and the given equation has one solution rather than infinitely many solutions.

Choice C is incorrect. The system consisting of this equation and the given equation has no solutions rather than infinitely many solutions.

Question Difficulty:
Medium

Question ID bd45df49

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: bd45df49

$$\begin{aligned}y &= 3x + 9 \\ 3y &= 8x - 6\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $x - y$?

- A. -123
- B. -33
- C. 3
- D. 57

ID: bd45df49 Answer

Correct Answer:
D

Rationale

Choice D is correct. The first equation in the given system of equations defines y as $3x + 9$. Substituting $3x + 9$ for y in the second equation in the given system of equations yields $3(3x + 9) = 8x - 6$. Applying the distributive property on the left-hand side of this equation yields $9x + 27 = 8x - 6$. Subtracting $8x$ from both sides of this equation yields $x + 27 = -6$. Subtracting 27 from both sides of this equation yields $x = -33$. Substituting -33 for x in the first equation of the given system of equations yields $y = 3(-33) + 9$, or $y = -90$. Substituting -33 for x and -90 for y into the expression $x - y$ yields $-33 - (-90)$, or 57 .

Choice A is incorrect. This is the value of $x + y$, not $x - y$.

Choice B is incorrect. This is the value of x , not $x - y$.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID d909cd31

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: d909cd31

$$-15x + 25y = 65$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. Which of the following could be the second equation in the system?

- A. $12x + 20y = 52$
- B. $12x + 20y = -52$
- C. $-12x + 20y = 52$
- D. $-12x + 20y = -52$

ID: d909cd31 Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions when the two linear equations are equivalent. Dividing both sides of the given equation by 5 yields $-3x + 5y = 13$. Dividing both sides of choice C by 4 also yields $-3x + 5y = 13$, so choice C is equivalent to the given equation. Thus, choice C could be the second equation in the system.

Choice A is incorrect. The system consisting of this equation and the given equation has one solution, not infinitely many solutions.

Choice B is incorrect. The system consisting of this equation and the given equation has one solution, not infinitely many solutions.

Choice D is incorrect. The system consisting of this equation and the given equation has no solution, not infinitely many solutions.

Question Difficulty:
Medium

Question ID 092ad67d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 092ad67d

$$\begin{aligned}x + 2y &= 6 \\ x - 2y &= 4\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. 2.5
- B. 5
- C. 6
- D. 10

ID: 092ad67d Answer

Correct Answer:
B

Rationale

Choice B is correct. Adding the first equation to the second equation in the given system yields $(x + 2y) + (x - 2y) = 6 + 4$, or $(x + x) + (2y - 2y) = 10$. Combining like terms in this equation yields $2x = 10$. Dividing both sides of this equation by 2 yields $x = 5$. Thus, the value of x is 5.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $2x$, not x .

Question Difficulty:
Medium

Question ID 79be5180

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 79be5180

$$\begin{aligned}y &= 9x + 12 \\ x + 7y &= 20\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of y ?

ID: 79be5180 Answer

Correct Answer:
3

Rationale

The correct answer is **3**. It's given that $y = 9x + 12$. Substituting $9x + 12$ for y in the second equation in the system, $x + 7y = 20$, yields $x + 7(9x + 12) = 20$, which gives $x + 63x + 84 = 20$, or $64x + 84 = 20$. Subtracting **84** from each side of this equation yields $64x = -64$. Dividing each side of this equation by **64** yields $x = -1$. Substituting -1 for x in the first equation in the system, $y = 9x + 12$, yields $y = 9(-1) + 12$, or $y = 3$. Therefore, the value of y is **3**.

Question Difficulty:
Medium

Question ID b00ad7f2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: b00ad7f2

$$\begin{aligned}y - 9x &= 13 \\ 5x &= 2y\end{aligned}$$

What is the solution (x, y) to the given system of equations?

- A. $(\frac{5}{2}, 1)$
- B. $(1, \frac{2}{5})$
- C. $(-2, -5)$
- D. $(-5, -2)$

ID: b00ad7f2 Answer

Correct Answer:
C

Rationale

Choice C is correct. Adding $9x$ to both sides of the first equation in the given system yields $y = 9x + 13$. Substituting the expression $9x + 13$ for y in the second equation in the given system yields $5x = 2(9x + 13)$. Distributing the 2 on the right-hand side of this equation yields $5x = 18x + 26$. Subtracting $18x$ from both sides of this equation yields $-13x = 26$. Dividing both sides of this equation by -13 yields $x = -2$. Substituting -2 for x in the equation $y = 9x + 13$ yields $y = 9(-2) + 13$, or $y = -5$. Therefore, the solution (x, y) to the given system of equations is $(-2, -5)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the solution (y, x) , not (x, y) , to the given system of equations.

Question Difficulty:
Medium

Question ID c5082ce3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: c5082ce3

The score on a trivia game is obtained by subtracting the number of incorrect answers from twice the number of correct answers. If a player answered 40 questions and obtained a score of 50, how many questions did the player answer correctly?

ID: c5082ce3 Answer

Rationale

The correct answer is 30. Let x represent the number of correct answers from the player and y represent the number of incorrect answers from the player. Since the player answered 40 questions in total, the equation $x + y = 40$ represents this situation. Also, since the score is found by subtracting the number of incorrect answers from twice the number of correct answers and the player received a score of 50, the equation $2x - y = 50$ represents this situation. Adding the equations in the system of two equations together yields $(x + y) + (2x - y) = 40 + 50$. This can be rewritten as $3x = 90$. Finally, solving for x by dividing both sides of the equation by 3 yields $x = 30$.

Question Difficulty:
Medium

Question ID a0489274

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: a0489274

$$\begin{aligned}y &= -\frac{1}{5}x \\ y &= \frac{1}{7}x\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. -5
- B. 0
- C. 2
- D. 7

ID: a0489274 Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given by the first equation in the system that $y = -\frac{1}{5}x$. Substituting $-\frac{1}{5}x$ for y in the second equation in the system, $y = \frac{1}{7}x$, yields $-\frac{1}{5}x = \frac{1}{7}x$. Adding $-\frac{1}{5}x$ to both sides of this equation yields $0 = \frac{1}{7}x + \frac{1}{5}x$, which is equivalent to $0 = \frac{5}{35}x + \frac{7}{35}x$, or $0 = \frac{12}{35}x$. Multiplying both sides of this equation by $\frac{35}{12}$ yields $0 = x$. Therefore, the value of x is 0 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 19fdf387

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 19fdf387

In the xy -plane, the graph of $y = x + 3$ intersects the graph of $y = 2x - 6$ at the point (a,b) . What is the value of a ?

- A. 3
- B. 6
- C. 9
- D. 12

ID: 19fdf387 Answer

Correct Answer:
C

Rationale

Choice C is correct. Since the graph of $y = x + 3$ intersects the graph of $y = 2x - 6$ at the point (a,b) , the ordered pair (a,b) is the solution to the system of linear equations consisting of $y = x + 3$ and $y = 2x - 6$, and the value of a is the value of x in the solution of this system. Since both $x + 3$ and $2x - 6$ are equal to y , it follows that $x + 3 = 2x - 6$. Subtracting x from and adding 6 to both sides of the equation yields $9 = x$. Therefore, the value of a is 9.

Choices A and B are incorrect and may result from a calculation or conceptual error in solving the system of equations consisting of $y = x + 3$ and $y = 2x - 6$. Choice D is incorrect. This is the value of b , not a .

Question Difficulty:
Medium

Question ID 6ba39da5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 6ba39da5

$$\begin{aligned}2a + 8b &= 198 \\ 2a + 4b &= 98\end{aligned}$$

The solution to the given system of equations is (a, b) . What is the value of b ?

ID: 6ba39da5 Answer

Correct Answer:
25

Rationale

The correct answer is **25**. Subtracting the second equation from the first equation in the given system of equations yields $(2a - 2a) + (8b - 4b) = 198 - 98$, which is equivalent to $0 + 4b = 100$, or $4b = 100$. Dividing each side of this equation by **4** yields $b = 25$.

Question Difficulty:
Medium

Question ID ed92fb68

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: ed92fb68

$$\begin{aligned}4x + 5y &= 100 \\ 5x + 4y &= 62\end{aligned}$$

If the system of equations above has solution (x, y) ,
what is the value of $x + y$?

- A. 0
- B. 9
- C. 18
- D. 38

ID: ed92fb68 Answer

Correct Answer:
C

Rationale

Choice C is correct. Adding the given equations yields $9x + 9y = 162$. Dividing each side of the equation $9x + 9y = 162$ by 9 gives $x + y = 18$.

Choice A is incorrect and may result from incorrectly adding the equations. Choice B is incorrect and may result from conceptual or computational errors. Choice D is incorrect. This value is equivalent to $y - x$.

Question Difficulty:
Medium

Question ID 9a216c0f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 9a216c0f

The combined original price for a mirror and a vase is **\$60**. After a **25%** discount to the mirror and a **45%** discount to the vase are applied, the combined sale price for the two items is **\$39**. Which system of equations gives the original price m , in dollars, of the mirror and the original price v , in dollars, of the vase?

- A. $m + v = 60$
 $0.55m + 0.75v = 39$
- B. $m + v = 60$
 $0.45m + 0.25v = 39$
- C. $m + v = 60$
 $0.75m + 0.55v = 39$
- D. $m + v = 60$
 $0.25m + 0.45v = 39$

ID: 9a216c0f Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that m represents the original price, in dollars, of the mirror, and v represents the original price, in dollars, of the vase. It's also given that the combined original price for the mirror and the vase is **\$60**. This can be represented by the equation $m + v = 60$. After a **25%** discount to the mirror is applied, the sale price of the mirror is **75%** of its original price. This can be represented by the expression $0.75m$. After a **45%** discount to the vase is applied, the sale price of the vase is **55%** of its original price. This can be represented by the expression $0.55v$. It's given that the combined sale price for the two items is **\$39**. This can be represented by the equation $0.75m + 0.55v = 39$. Therefore, the system of equations consisting of the equations $m + v = 60$ and $0.75m + 0.55v = 39$ gives the original price m , in dollars, of the mirror and the original price v , in dollars, of the vase.

Choice A is incorrect. The second equation in this system of equations represents a **45%** discount to the mirror and a **25%** discount to the vase.

Choice B is incorrect. The second equation in this system of equations represents a **55%** discount to the mirror and a **75%** discount to the vase.

Choice D is incorrect. The second equation in this system of equations represents a **75%** discount to the mirror and a **55%** discount to the vase.

Question Difficulty:
Medium

Question ID 8a87c2c8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 8a87c2c8

$$\begin{aligned}x + 3 &= -2y + 5 \\ x - 3 &= 2y + 7\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $2x$?

- A. -2
- B. 6
- C. 12
- D. 24

ID: 8a87c2c8 Answer

Correct Answer:
C

Rationale

Choice C is correct. Adding the second equation in the given system to the first equation in the given system yields $(x + 3) + (x - 3) = (-2y + 5) + (2y + 7)$. Adding like terms in this equation yields $2x = 12$. Thus, the value of $2x$ is 12 .

Choice A is incorrect. This is the value of y , not $2x$.

Choice B is incorrect. This is the value of x , not $2x$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 74c03c21

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 74c03c21

A bus traveled on the highway and on local roads to complete a trip of **160 miles**. The trip took **4 hours**. The bus traveled at an average speed of **55 miles per hour (mph)** on the highway and an average speed of **25 mph** on local roads. If x is the time, in hours, the bus traveled on the highway and y is the time, in hours, it traveled on local roads, which system of equations represents this situation?

- A. $55x + 25y = 4$
 $x + y = 160$
- B. $55x + 25y = 160$
 $x + y = 4$
- C. $25x + 55y = 4$
 $x + y = 160$
- D. $25x + 55y = 160$
 $x + y = 4$

ID: 74c03c21 Answer

Correct Answer:
B

Rationale

Choice B is correct. If the bus traveled at an average speed of **55 miles per hour (mph)** on the highway for x hours, then the bus traveled **$55x$ miles** on the highway. If the bus traveled at an average speed of **25 mph** on local roads for y hours, then the bus traveled **$25y$ miles** on local roads. It's given that the trip was **160 miles**. This can be represented by the equation **$55x + 25y = 160$** . It's also given that the trip took **4 hours**. This can be represented by the equation **$x + y = 4$** . Therefore, the system consisting of the equations **$55x + 25y = 160$** and **$x + y = 4$** represents this situation.

Choice A is incorrect. This system of equations represents a situation where the trip was **4 miles** and took **160 hours**.

Choice C is incorrect. This system of equations represents a situation where the trip was **4 miles** and took **160 hours**, and the bus traveled at an average speed of **25 mph** on the highway and **55 mph** on local roads.

Choice D is incorrect. This system of equations represents a situation where the bus traveled at an average speed of **25 mph** on the highway and **55 mph** on local roads.

Question Difficulty:
Medium

Question ID f5929f7a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: f5929f7a

$$\begin{aligned}y &= -\frac{1}{9}x \\ y &= \frac{1}{2}x\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. -9
- B. -7
- C. 0
- D. 2

ID: f5929f7a Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given by the first equation in the system that $y = -\frac{1}{9}x$. Substituting $-\frac{1}{9}x$ for y in the second equation in the system yields $-\frac{1}{9}x = \frac{1}{2}x$. Multiplying the left-hand side of this equation by $\frac{2}{2}$ and the right-hand side by $\frac{9}{9}$ yields $-\frac{2}{18}x = \frac{9}{18}x$. Adding $\frac{2}{18}x$ to both sides of this equation yields $0 = \frac{11}{18}x$. Multiplying both sides of this equation by $\frac{18}{11}$ yields $x = 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 6e6a3241

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 6e6a3241

$x + 5y = 5$
 $2x - y = -4$

Which of the following graphs in the xy -plane could be used to solve the system of equations above?

- A.
- B.
- C.
- D.

ID: 6e6a3241 Answer

Correct Answer:
C

Rationale

Choice C is correct. The graph of a system of equations is the graph that shows the lines represented by each of the equations in the system. The x -intercept of the graph of each given equation can be found by substituting 0 for y in each equation: $x + 5(0) = 5$, or $x = 5$, and $2x - 0 = -4$, or $x = -2$. The y -intercept of the graph of each equation can be found by substituting 0 for x in each equation: $0 + 5y = 5$, or $y = 1$, and $2(0) - y = -4$ or $y = 4$. Using these x - and y -intercept values, the line that has equation $x + 5y = 5$ passes through the points $(0, 1)$ and $(5, 0)$, and the line that has equation $2x - y = -4$ passes through the points $(0, 4)$ and $(-2, 0)$. Only the lines in choice C pass through these points and can be used to solve the given system of equations.

Choices A, B, and D are incorrect. In choices A and B, neither line passes through $(0,1)$ and $(5,0)$ or $(0,4)$ and $(-2,0)$. In choice D, although one line passes through $(0,4)$ and $(-2,0)$ the other line doesn't pass through $(0,1)$ and $(5,0)$.

Question Difficulty:

Medium

Question ID 6a87902f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 6a87902f

$y = 2x + 10$
 $y = 2x - 1$

At how many points do the graphs of the given equations intersect in the xy -plane?

- A. Zero
- B. Exactly one
- C. Exactly two
- D. Infinitely many

ID: 6a87902f Answer

Correct Answer:
A

Rationale

Choice A is correct. A system of two linear equations in two variables, x and y , has zero points of intersection if the lines represented by the equations in the xy -plane are distinct and parallel. The graphs of two lines in the xy -plane represented by equations in slope-intercept form, $y = mx + b$, are distinct if the y -coordinates of their y -intercepts, b , are different and are parallel if their slopes, m , are the same. For the two equations in the given system, $y = 2x + 10$ and $y = 2x - 1$, the values of b are 10 and -1 , respectively, and the values of m are both 2 . Since the values of b are different, the graphs of these lines have different y -coordinates of the y -intercept and are distinct. Since the values of m are the same, the graphs of these lines have the same slope and are parallel. Therefore, the graphs of the given equations are lines that intersect at zero points in the xy -plane.

Choice B is incorrect. The graphs of a system of two linear equations have exactly one point of intersection if the lines represented by the equations have different slopes. Since the given equations represent lines with the same slope, there is not exactly one intersection point.

Choice C is incorrect. The graphs of a system of two linear equations can never have exactly two intersection points.

Choice D is incorrect. The graphs of a system of two linear equations have infinitely many intersection points when the lines represented by the equations have the same slope and the same y -coordinate of the y -intercept. Since the given equations represent lines with different y -coordinates of their y -intercepts, there are not infinitely many intersection points.

Question Difficulty:
Medium

Question ID 38a43902

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 38a43902

$$\begin{aligned}y &= -2x \\ 3x + y &= 40\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

ID: 38a43902 Answer

Correct Answer:
40

Rationale

The correct answer is **40**. It's given in the first equation of the system that $y = -2x$. Substituting $-2x$ for y in the second equation of the system yields $3x + (-2x) = 40$. Combining like terms on the left-hand side of this equation yields $x = 40$. Therefore, the value of x is **40**.

Question Difficulty:
Medium

Question ID 0c541d87

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 0c541d87

Two customers purchased the same kind of bread and eggs at a store. The first customer paid **12.45** dollars for **1** loaf of bread and **2** dozen eggs. The second customer paid **19.42** dollars for **4** loaves of bread and **1** dozen eggs. What is the cost, in dollars, of **1** dozen eggs?

- A. **3.77**
- B. **3.88**
- C. **4.15**
- D. **4.34**

ID: 0c541d87 Answer

Correct Answer:
D

Rationale

Choice D is correct. Let ℓ represent the cost, in dollars, of **1** loaf of bread, and let d represent the cost, in dollars, of **1** dozen eggs. It's given that the first customer paid **12.45** dollars for **1** loaf of bread and **2** dozen eggs. Therefore, the first customer's purchase can be represented by the equation $\ell + 2d = 12.45$. It's also given that the second customer paid **19.42** dollars for **4** loaves of bread and **1** dozen eggs. Therefore, the second customer's purchase can be represented by the equation $4\ell + d = 19.42$. The equations $\ell + 2d = 12.45$ and $4\ell + d = 19.42$ form a system of linear equations, which can be solved by elimination to find the value of d . Multiplying the first equation in the system by -4 yields $-4\ell - 8d = -49.8$. Adding $-4\ell - 8d = -49.8$ to the second equation, $4\ell + d = 19.42$, yields $(-4\ell + 4\ell) + (-8d + d) = (-49.8 + 19.42)$, which is equivalent to $-7d = -30.38$. Dividing both sides of this equation by -7 yields $d = 4.34$. Therefore, the cost, in dollars, of **1** dozen eggs is **4.34**.

Choice A is incorrect. This is the cost, in dollars, of **1** loaf of bread.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 71189542

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 71189542

A group of 202 people went on an overnight camping trip, taking 60 tents with them. Some of the tents held 2 people each, and the rest held 4 people each. Assuming all the tents were filled to capacity and every person got to sleep in a tent, exactly how many of the tents were 2-person tents?

- A. 30
- B. 20
- C. 19
- D. 18

ID: 71189542 Answer

Correct Answer:
C

Rationale

Choice C is correct. Let x represent the number of 2-person tents and let y represent the number of 4-person tents. It is given that the total number of tents was 60 and the total number of people in the group was 202. This situation can be expressed as a system of two equations, $x + y = 60$ and $2x + 4y = 202$. The first equation can be rewritten as $y = -x + 60$. Substituting $-x + 60$ for y in the equation $2x + 4y = 202$ yields $2x + 4(-x + 60) = 202$. Distributing and combining like terms gives $-2x + 240 = 202$. Subtracting 240 from both sides of $-2x + 240 = 202$ and then dividing both sides by -2 gives $x = 19$. Therefore, the number of 2-person tents is 19.

Alternate approach: If each of the 60 tents held 4 people, the total number of people that could be accommodated in tents would be 240. However, the actual number of people who slept in tents was 202. The difference of 38 accounts for the 2-person tents. Since each of these tents holds 2 people fewer than a 4-person tent, $\frac{38}{2} = 19$ gives the number of 2-person tents.

Choice A is incorrect. This choice may result from assuming exactly half of the tents hold 2 people. If that were true, then the total number of people who slept in tents would be $2(30) + 4(30) = 180$; however, the total number of people who slept in tents was 202, not 180. Choice B is incorrect. If 20 tents were 2-person tents, then the remaining 40 tents would be 4-person tents. Since all the tents were filled to capacity, the total number of people who slept in tents would be $2(20) + 4(40) = 40 + 160 = 200$; however, the total number of people who slept in tents was 202, not 200. Choice D is incorrect. If 18 tents were 2-person tents, then the remaining 42 tents would be 4-person tents. Since all the tents were filled to capacity, the total number of people who slept in tents would be $2(18) + 4(42) = 36 + 168 = 204$; however, the total number of people who slept in tents was 202, not 204.

Question Difficulty:

Question ID 7efe5495

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 7efe5495

$y = 3x$
 $2x + y = 12$

The solution to the given system of equations is (x, y) . What is the value of $5x$?

- A. 24
- B. 15
- C. 12
- D. 5

ID: 7efe5495 Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given by the first equation in the system that $y = 3x$. Substituting $3x$ for y in the equation $2x + y = 12$ yields $2x + 3x = 12$, or $5x = 12$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID Odd6227f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: Odd6227f

At how many points do the graphs of the equations $y = x + 20$ and $y = 8x$ intersect in the xy -plane?

- A. 0
- B. 1
- C. 2
- D. 8

ID: Odd6227f Answer

Correct Answer:

B

Rationale

Choice B is correct. Each given equation is written in slope-intercept form, $y = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the graph of the equation in the xy -plane. The graphs of two lines that have different slopes will intersect at exactly one point. The graph of the first equation is a line with slope 1. The graph of the second equation is a line with slope 8. Since the graphs are lines with different slopes, they will intersect at exactly one point.

Choice A is incorrect because two graphs of linear equations have 0 intersection points only if they are parallel and therefore have the same slope.

Choice C is incorrect because two graphs of linear equations in the xy -plane can have only 0, 1, or infinitely many points of intersection.

Choice D is incorrect because two graphs of linear equations in the xy -plane can have only 0, 1, or infinitely many points of intersection.

Question Difficulty:

Medium